REACTIVEDYNAMICS.JL — THE AGENTIC LAYER

ReactiveDynamics.jl Modeling Copilot

From a capacity-constrained business decision to a checked, reproducible model and a number you can defend, in a conversation, without anyone writing code.

Capacity decisions need a model, and a model needs a modeler

- A launch reservation, a bid under a staffing ceiling, a portfolio competing for the same people: the answer depends on **who waits when the line is full**.
- A spreadsheet cannot represent a queue. Contention is an interaction between rows, and a spreadsheet's rows are independent by construction.
- So these questions go to a specialist, take weeks, and come back as a workbook the decision-maker has to take on faith.
- Most never get a model at all.

THE VALUE

More decisions get a real model, and the owner stays in the loop

LINES OF CODE YOU WRITE

0

the model is a checked document authored in conversation

GATED PHASES

4

frame, build, try to break it, report

PLAUSIBILITY CHECKS

9

a failure blocks the headline number

SEED REPLAYS IT

1

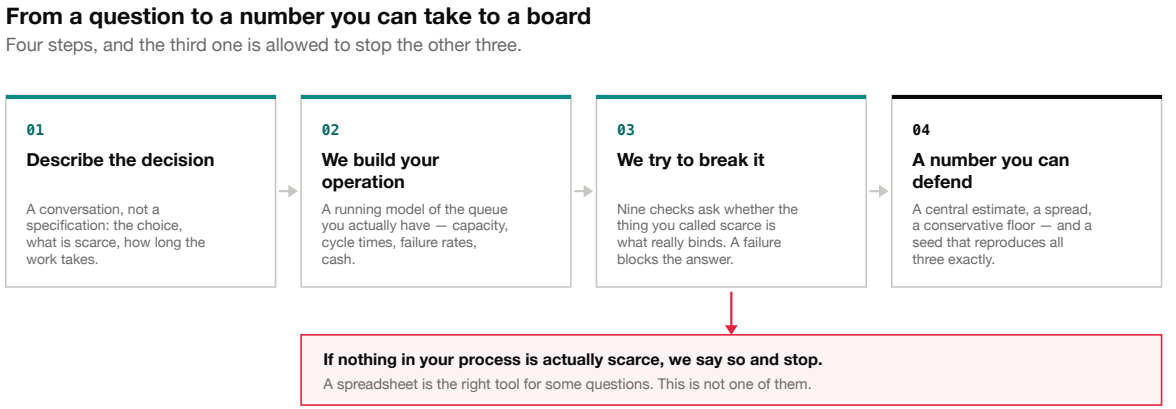
same model, same seed, same answer

The interview runs in business language with the person who owns the decision. Every assumption lands in a brief they signed, including which inputs were their own estimates. The next question about the same operation starts from the same model.

HOW IT WORKS

Four steps, and the third may stop the rest

Step	You get	Gate
Frame	a signed brief	something must bind
Author	a checked model per option	validator silent
Diagnose	a health report	red check blocks
Report	estimate, mechanism, evidence	every number has a seed



The model is data, not code, so it can be checked before it runs

Static checks. A pure validator reads the model document and returns every problem with its exact path. A malformed model produces diagnostics, not a plausible wrong number. The language model proposes and a deterministic checker disposes, and the loop repeats until the checker is silent.

Nothing generated is executed. No `eval`, no generated scripts. Custom logic is one declared file.

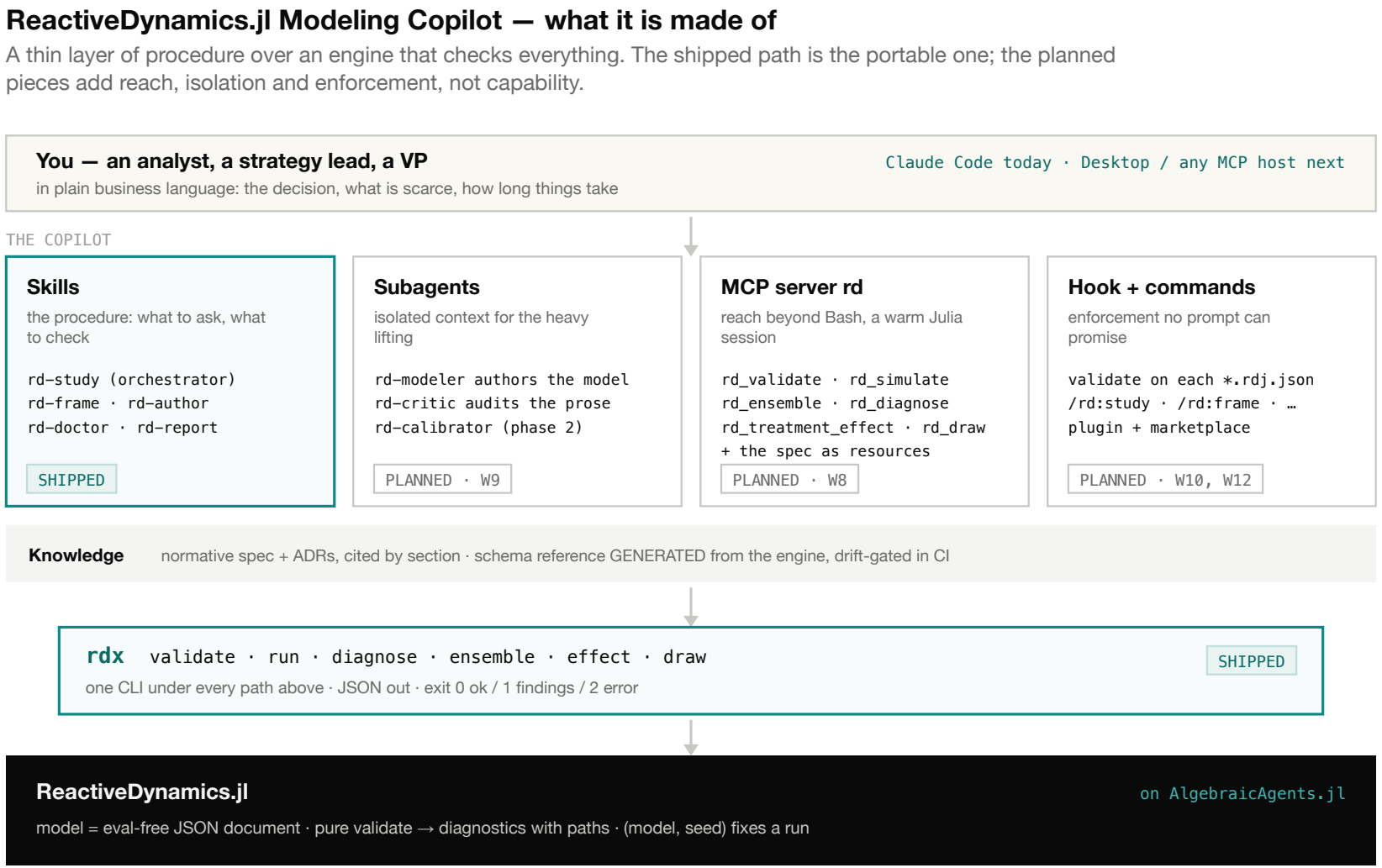
The IR. The model is a closed, eval-free document: places that hold resources, transitions that hold them over time and sometimes fail, rates, costs, the arm deltas. It is diffable, serializable and replayable from `(model, seed)`.

Checked again, for meaning. Nine plausibility checks ask whether the scarce thing actually binds and whether the horizon cuts off the payoff.

Guarded in CI. The schema reference is generated from the engine, and a known-answer test re-derives a published result on every change.

WHAT IT IS MADE OF

A thin layer over an engine that checks everything



A REAL SESSION

Launch supply: do we pay for capacity before the readouts?

Blind-captured: two agents that cannot see each other's reasoning. The operator wrote its facts down before the first message. The company is invented, and the numbers are measurements.

Two Phase-3 biologics, one fill-finish line, a CMO offer that expires 30 November

OPERATOR

Two Phase-3 biologics are coming. […] We have one sterile fill-finish line in-house, and most of it is already taken by our marketed products. It can probably carry one launch but not two. […] Our CFO says we shouldn't pay for capacity on an asset that has less than even odds.

CLAUDE

The CMO slot works for Tern's filing, but only just. If you sign by 30 November, PPQ is done somewhere between late 2027 and early 2028. […] So the CMO can be Tern's filed site with no PAS, but only if the CMO runs on time and Tern doesn't read out early.

```
python3 -c '<in-house headroom: 12.8 batches/mo minus marketed 9/mo growing 5%/yr>'
```

One model, an arm per option, calibrated to the operator's own figures

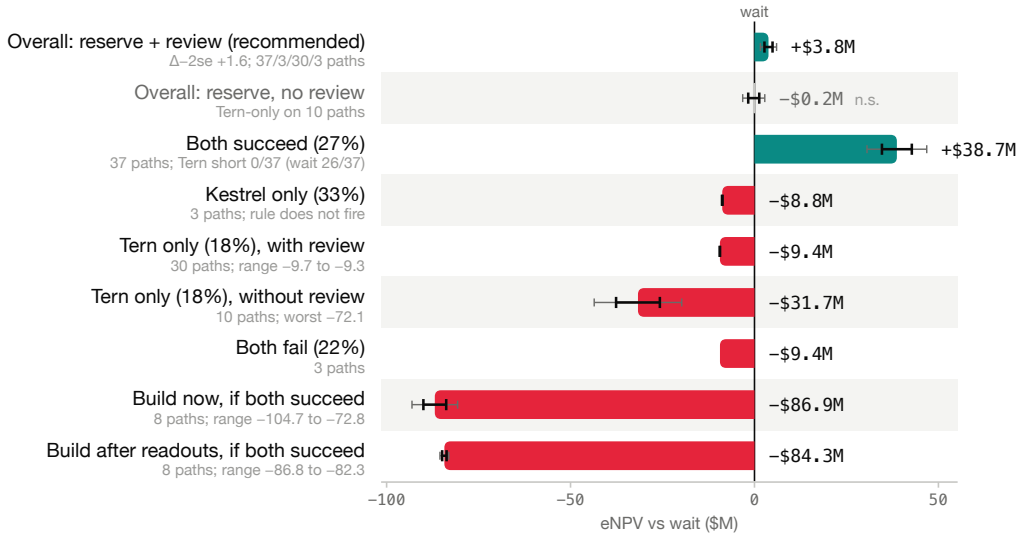
- `gen_models.py` writes one network; the arms differ only in parameters.
- `rdx validate`: 0 errors on every document.
- Calibration: Kestrel first-year revenue **$250.0M** against the operator's ~$250M, and **54** batches/yr at peak against 54.
- `rdx diagnose`: the own line binds (128 of 128 tenths used), and the design-dead branches are explained in `DOCTOR.md`.
- Options compared on **paired paths**: the same seed gives every option the same readout dates and lead times.
- Readout outcomes are forced per path and weighted by the stated odds (0.27 / 0.33 / 0.18 / 0.22).
- No option-level ensembles were run: a 96-seed ensemble took 53 minutes (W24). One wait ensemble served as a cross-check, and replays of three of its members matched exactly.

Reserve now, review at Kestrel's readout, don't build

- **+$3.8M ± $1.1M** against waiting, odds-weighted; +$25M at peak-year batch value.
- **+$39M if both succeed** (27%): Tern short at launch on 26 of 37 paths under wait, 0 of 37 reserved.
- **A bounded $9–11M cost** in every other outcome.
- **Building is $73–105M worse** even when both succeed.
- The operator's prior, a Tern-only reservation, leaves Kestrel short on every both-succeed path.

Reserve the CMO now, review at Kestrel's readout: eNPV against waiting

Paired single paths (same seed, same dates in both options), whiskers ±1 se (dark) and ±2 se (grey) across paths. Top two bars: odds-weighted overall. Others: given the outcome. Kestrel 60%, Tern 45%, 9% WACC, year-1 value per batch.



THE MOMENT WORTH WATCHING

The model made the tail visible, and the owner supplied the option

CLAUDE

In the other outcomes the reservation costs money: Tern only, 0.18 chance, **–$18.6M to –$72.1M**. […] That is why the expected value nets to about zero.

OPERATOR

Kestrel reads out in March 2027, nine months before Tern. […] I'd move Tern's PPQ in-house, file Tern on our own line, and give the CMO notice. That means eating the $5–10M. So the reservation isn't a fixed bet on Tern; it's something we decide to keep or drop at Kestrel's readout.

CLAUDE

Your guess holds: under the review rule, the Tern-only outcome costs $9.4M against waiting, and the $72M tail is gone.

```
python3 exit_review.py → weighted delta +3.81, se 1.10 (30 Tern-only paths)
```

HONEST ABOUT IT

What we are not claiming

- **The case is simulated.** Two role-playing agents; the operator was cooperative. It demonstrates the workflow, not anyone's supply plan.
- **Paired paths, not ensembles.** The run budget ruled ensembles out, and every figure says what it rests on.
- **The interview left soft inputs untested:** demand ±30%, marketed growth and readout correlation. The published facts audit leads with them.
- **The planned components are planned.** MCP, subagents, the hook and the plugin follow endorsement.

THE ASK

Endorse the capability; we build the reach

Next	What it adds	Tracker
MCP server `rd`	the six operations as tools in any MCP host, no terminal	W8
Subagents	an isolated author and a critic that audits the prose against the numbers	W9
Validate hook	every model edit validated automatically	W10
Study card	a one-page shareable summary	W11
Plugin	one-step install and slash commands	W12

The full argument, with the transcript and its evidence, is `agent/copilot.html`.